THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\sqrt{484}$ is equal to: 1

(a) 24 **(b)** 21 **(c)** 22 **(d)** 23

2. The value of 8^3 is: 1

(a) 24 **(b)** 512 **(c)** 192 **(d)** 520

3. $\sqrt[3]{343}$ is: 1

(a) 6 **(b)** 7 **(c)** 9 **(d)** 8

4. The sum of the first 10 odd natural numbers is: 1

(a) 20 **(b)** 100 **(c)** 99 **(d)** 110

5. Which of the following is a perfect square? 1

(a) 121 **(b)** 127 **(c)** 123 **(d)** 118

6. The value of 21^2 is: 1

(a) 462 **(b)** 420 **(c)** 441 **(d)** 42

7. $\sqrt{576}$ is equal to: 1

(a) 26 **(b)** 25 **(c)** 24 **(d)** 23

8. The value of 10^3 is: 1

(a) 1010 **(b)** 30 **(c)** 1000 **(d)** 300

9. $\sqrt[3]{512}$ is: 1

(a) 9 **(b)** 10 **(c)** 7 **(d)** 8

10. Which of the following is a perfect square? 1

(a) 144 **(b)** 141 **(c)** 146 **(d)** 150

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of 16^3. 2

12. Find the value of 52^2. 2

13. Is 500 a perfect square? Give a reason. 2

14. Find the value of 20^3. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find $\sqrt{784}$ by prime factorisation. 3

16. Find $\sqrt[3]{2744}$ by prime factorisation. 3

17. Write 121 as the sum of the first 11 odd natural numbers (show the sum pattern). 3

OR

Find $\sqrt{900}$ by prime factorisation. 3

18. Find $\sqrt[3]{3375}$ by prime factorisation. 3

OR

Write 144 as the sum of the first 12 odd natural numbers (show the sum pattern). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the smallest number by which 490 must be multiplied so that the product is a perfect square. Also find the square root of the product. 5

OR

Find the smallest number by which 2000 must be divided so that the quotient is a perfect cube. Also find the cube root of the quotient. 5

20. Find the smallest number by which 588 must be multiplied so that the product is a perfect square. Also find the square root of the product. 5

OR

Find the smallest number by which 3240 must be divided so that the quotient is a perfect cube. Also find the cube root of the quotient. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND